Lecture 4: Multiparameter Models

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2026-01-12

Announcements

- Today: Chapter 3
- Next: Chapter 4 (can skim) and Chapter 5

(Hierarchical Modeling)

- Hwk due Sunday midnight.

Bayesian inference in the normal model

$=1$ e.g.

- Assume $y_1, \ldots y_n \sim N(\mu, \sigma^2)$ with σ^2 a known constant
- Let's start with a Jeffreys' (improper prior): $p(\mu) \propto \text{const}$
- What is the posterior distribution $p(\mu \mid y_1, \ldots y_n, \sigma^2)$?

Jeff. $\sqrt{I(\mu)} = \sqrt{\frac{n}{\sigma^2}} \propto$ const. (improper!)

$$P(\mu | y_1, \ldots y_n) \propto L(\mu) P(\mu) \propto L(\mu)$$

$$\propto \prod_{i=1}^{n} e^{-\frac{(y_i-\mu)^2}{2\sigma^2}} \propto e^{-\frac{\sum(y_i-\mu)^2}{2\sigma^2}}$$

$$\propto e^{\frac{-n\mu^2+2(\sum y_i)\mu - \sum y_i^2}{2\sigma^2}}$$

$1/n$

$1/n$

$$\propto e^{\frac{-\mu^2+2\bar{y}\mu}{2\sigma^2/n}} \times \cancel{e^{\frac{\sum y_i^2}{n}}}$$

$$\propto e^{\frac{-(\mu-\bar{y})^2}{2\sigma^2/n}} \times \text{const}$$

<u>Bayes</u> $$\mu \mid y_1, \dots y_n, \sigma^2 \sim N\Big(\overbrace{\bar{y}}^{E[\mu \mid y_1, \dots y_n]}, \underbrace{\frac{\sigma^2}{n}}_{Var(\mu \mid y_1, \dots y_n)}\Big)$$

<u>Freq.</u>

$$\hat{\mu}_{MLE} = \bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\sqrt{\sigma^2/n}\,(\mu - \bar{y}) \sim N(0, 1)$$

$$E[\mu \mid y_1, \dots y_n] = \bar{y} = \hat{\mu}_{MLE}$$

Confidence Interval equiv. Bayes Credible
Interpretable.

Bayesian inference in the normal model

- Assume $y_1, \ldots y_n \sim N(\mu, \sigma^2)$ with σ^2 a known constant
- The normal prior distribution is conjugate for μ in the normal sampling model
- Sampling distribution, prior distribution and posterior distribution are all normal.
- Assume the prior is $p(\mu) \sim N(\mu_0, \underbrace{\sigma^2/\kappa_0}_{\tau^2})$
- What are the parameters of the posterior $p(\mu \mid y_1, \ldots y_n, \sigma^2)$?

Prior Parameter

Normal-Normal model.

$$P(\mu \mid y_1, \dots y_n, \sigma^2) \propto L(\mu) P(\mu)$$

$$\propto \underbrace{e^{-\frac{(\mu-\bar{y})^2}{2\sigma^2/n}}}_{L(\mu)} \underbrace{e^{-\frac{(\mu-\mu_0)^2}{2\sigma^2/k_0}}}_{P(\mu)}$$

$$\propto \exp\left[-\frac{1}{2}\left(\left(\frac{n+k_0}{\sigma^2}\right)\mu^2 - 2\left(\frac{n\bar{y}}{\sigma^2} + \frac{k_0}{\sigma^2}\mu_0\right)\mu + \text{consts}\right]\right.$$

$$\propto \exp\left[-\frac{1}{2}\left(\frac{n+k_0}{\sigma^2}\right)\left(\mu - \frac{\frac{n\bar{y}}{\sigma^2} + \frac{k_0}{\sigma^2}\mu_0}{\frac{n+k_0}{\sigma^2}}\right)^2\right]$$

$$\boxed{\mu \mid y_1, \dots y_n \sim N\left(w\bar{y} + (1-w)\mu_0, \frac{\sigma^2}{n+k_0}\right)}$$

$$E[\mu \mid y_1, \dots y_n] = \underset{\substack{\uparrow \\ \text{MLE}}}{w\bar{y}} + (1-w)\underset{\substack{\uparrow \\ \text{prior mean}}}{\mu_0}$$

$$w = \frac{\frac{n}{\sigma^2}}{\frac{n+k_0}{\sigma^2}} = \frac{n}{n+k_0}$$

"pseudo" sample size

"Total" = pseudo + Actual sample size.

$$Var(\mu \mid y_1, \dots y_n) = \frac{\sigma^2}{n+k_0}$$

A conjugate prior for the normal likelihood

- The normal distribution is conjugate for the normal likelihood
  - Often called the “normal-normal model”
- $Y_i \sim N(\mu, \sigma^2)$ and $\mu \sim N(\mu_0, \sigma^2/\kappa_0)$ implies that the posterior distribution $p(\mu \mid y)$ is also normally distributed:

$E[\mu | y_1, ... y_n]$

$$\mu \mid Y \sim N(\mu_n, \tau_n^2)$$

$Var(\mu | y_1, ... y_n)$

where $\mu_n = \dfrac{\frac{\kappa_0}{\sigma^2}\mu_0 + \frac{n}{\sigma^2}\bar{y}}{\frac{\kappa_0}{\sigma^2} + \frac{n}{\sigma^2}}$ and $\tau_n^2 = \dfrac{1}{\frac{\kappa_0}{\sigma^2} + \frac{n}{\sigma^2}}$

The posterior mean and pseudo-counts

$$\mu_n = \frac{\frac{1}{\tau^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}\mu_0 + \frac{\frac{n}{\sigma^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}\bar{y}$$

Fisher weight $= (1 - w)\mu_0 + w\bar{y}$

Fisher Information.

where $w = \frac{\frac{n}{\sigma^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}$

Prior variance: $\tau^2 = \frac{\sigma^2}{K_0}$

Prior precision

Precision-weighted.

Known mean, unknown variance

- Assume we have n mean-zero normal observations with variance σ^2

known

- Define $d_i = (y_i - \mu)$ for notational convenience
- What is $p(\sigma^2 \mid \mu, d_1, \ldots d_n)$?

$$L(\sigma^2) = \prod_{i=1}^{n} P(y_i \mid \sigma^2) = \prod \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\left(\frac{d_i^2}{2\sigma^2}\right)}$$

$$L\left(\sigma^2\right) \propto \left(\sigma^2\right)^{-n/2} \exp\left\{-\sum_{i=1}^{n} d_i^2 / \left(2\sigma^2\right)\right\}, \ \sigma^2 > 0$$

$$P(\sigma^2) \propto \frac{1}{\sigma^2}$$

Jeffreys:

$$P(\sigma^2) \propto 1/\sigma^2 \qquad \int_0^\infty \frac{1}{x}\,dx = \infty$$

$$P(\sigma) \propto 1/\sigma \qquad \text{(improper!)}$$

$$P(\log\sigma) \propto \text{const.}$$

Known mean, unknown variance

- Assume we have n mean-zero normal observations with variance σ^2
- Jeffreys prior $p(\sigma^2) \propto \frac{1}{\sigma^2}$ (careful!)
- The posterior:

$$p(\sigma^2 \mid y) \propto (\sigma^2)^{-n/2-1} \exp\left\{-\sum_{i=1}^n d_i^2 / (2\sigma^2)\right\}$$

- This distribution called an *inverse-Gamma* distribution.
  - If $X \sim \text{Gamma}(a, b)$ then $\frac{1}{X} \sim \text{Inv-Gamma}(a, b)$.

$z \sim$ Inv-Gam: $\quad z^{-} \; e^{-\frac{}{z}}$

$\sigma^2 / y_1, \ldots, y_n, \mu : \quad \text{Inv-Gamma}\left(n/2, \frac{\sum d_i^2}{2}\right)$

A conjugate prior distribution

- In general, the conjugate prior distribution has the form:

$$p(\sigma^2) \propto (\sigma^2)^{-k_1} e^{\frac{k_2}{\sigma^2}}$$

(I.G. Conjugate Prior)

- Inverse-gamma: $p(y \mid a, b) = \frac{b^a}{\Gamma(a)} y^{-a-1} \exp\left(-\frac{b}{y}\right)$
  - The mean of an inverse-Gamma is b/(a-1)

→ I.G. Posterior.

Scaled-Inverse-$\chi^2\left(n, \frac{\sum d_i^2}{n}\right)$ (BDA3)

$\Longleftrightarrow$

$I.G.\left(n/2, \frac{\sum d_i^2}{2}\right)$

$\chi_K^2 = \sum^K z_i^2, \quad z_i \sim N(0,1)$ (Chi-Squared)

$\sigma^2 \chi_K^2 \sim$ scaled-Chi-Sq

$\chi_K^2 \equiv \text{Gamma}\left(\frac{K}{2}, 1/2\right)$

$\left(\chi_K^2\right)^{-1} \equiv$ Inv.-Gamma.

Joint inference for the mean and variance

In the normal model we ~~typically factorize~~ the prior distribution $p(\mu, \sigma^2) = \boxed{p(\mu \mid \sigma^2)p(\sigma^2)}.$ $= p(\sigma^2/\mu)\,p(\mu)$

Normal

Inv-Gamma

Specifically:

$$\sigma^2 \sim Inv - Gamma(\nu_0/2, \nu_0/2\sigma_0^2)$$

$$\mu|\sigma^2 \sim \text{ normal } (\mu_0, \sigma^2/\kappa_0)$$

$$Y_1, \ldots, Y_n|\mu, \sigma \sim \text{ i.i.d. normal } (\mu, \sigma^2)$$

"Normal-inverse Gamma"

prior sample size for μ.

prior mean for μ

- ν_0 is interpreted as the prior sample size for σ^2
- σ_0^2 is a prior sample variance for σ^2